Cross-Layer Encryption of CFB-AES-TURBO for Advanced Satellite Data Transmission Security

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As the advanced satellite network can provide extremely diverse data to various types of users, it is of vital importance to achieve data security using encryption over satellite channels. However, since the avalanche effect makes encryption techniques vulnerable to bit errors in the wireless channel, a more powerful encryption technique is required for data confidentiality. In addition, an efficient method for exchanging encryption keys between receivers and transmitters should be capable of managing a large scale of satellites in the network. In this article, we propose a novel superencryption method of joint encryption and channel coding, cipher feedback-advanced encryption standard (CFB-AES)-TURBO, which combines AES in the upper layer and turbo coding as bulk encryption in the physical layer. Because we introduce turbo coding in the CFB mode with AES for each block, our scheme can accomplish dual goals of data protection and bit error correction. We select an encryption block size to maximize throughput based on channel conditions and the security level required for a user, and present a key exchanging solution. We derive processing time gains and analyze computational complexity of the CFB-AES-TURBO. The bit error rate performance improvement with about 12-dB coding gain is achieved. We also enhance the security performance by 10^{15000} times and provide indistinguishability. Finally, we analyze the total expected latency, suggest the minimum key refresh time of geostationary Earth orbit satellites.

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I. INTRODUCTION

Satellite communications have the advantages that they enable TV and audio broadcasting in a global scale, and they support mobile users in ships, aircrafts, vehicles, and high-speed trains. They also provide a wide range of data, such as voice, text, images, video streaming, emergency/disaster alarms, and so forth. As the number of users is rapidly increasing, data security is becoming a major issue in satellite networks [2]. IOActive, a research company for security services, reports security risks in aviation, maritime, military, and space applications exploiting satellite communications [3]. It warns of attacks against user devices connected to satellite communication equipment. Furthermore, the challenge of long latency over satellite channels [4] leads to easy exposure to physical layer attacks [5], for a long time in comparison with terrestrial networks. To solve this problem, in the physical layer, an information theoretic approach called physical layer security (PLS) has been studied in terrestrial and satellite networks [6]–[8]. However, this approach depends on the eavesdropper's (Eve) condition. For example, if Eve's channel condition is good or Eve has decoding (DEC) capability for encoded satellite confidential data, this will appreciably degrade the secrecy capacity, the difference between capacities of a legitimate user and Eve. Also, it is practically difficult for the satellite to obtain the channel condition of Eve. An alternative solution is that encryption techniques should be applied to data. Recently, several studies on security issues of satellite communication networks have been published [9]–[11]. Data confidentiality has been emphasized to protect information from unauthorized users (Eve or other users) due to the broadcasting nature. To use the encryption technique, one of the most important steps is a secret key exchange, which is the initial point to preserve data. Efficient key exchange methods have been investigated to manage a large volume of encryption keys.

The advanced encryption standard (AES) is the most prevalent block cipher encryption technique. AES converts a plaintext of 128 bits into a ciphertext of 128 bits using nonlinear mapping, called S-box, and a key [12]. In the process of encryption, each round replaces data with another character through four transformations. If the original data or the encryption key has a single bit change, the avalanche effect alters about half of the original data, and thus, makes AES more secure [13]. In a wireless system, however, Fig. 1 shows that the avalanche effect can be harmful because bit errors from channel fading can pollute the original data [14]. This results in increments of 0.4020 and 0.0627 between error probabilities before and after decryption, especially

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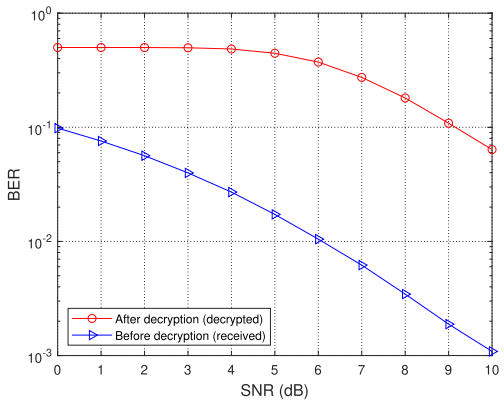


Fig. 1. Vulnerability of AES by the avalanche effect in the wireless system.

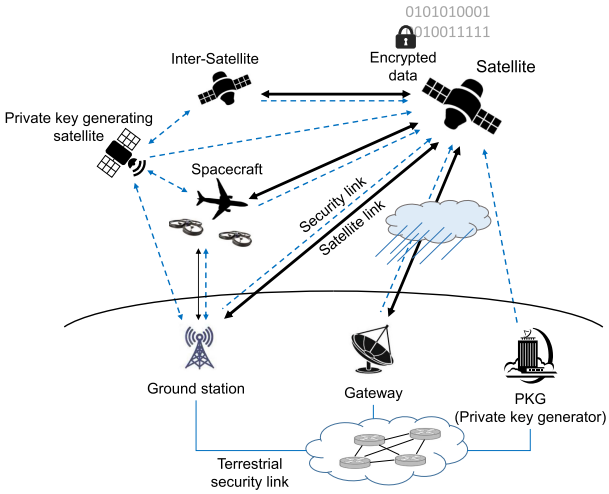


Fig. 2. Security system architecture of satellite communications.

in the signal-to-noise ratios (SNRs) of 0 and 10 dB, which implies that even a few bit errors can deteriorate original data severely. In addition, AES is exposed to cryptanalysis, such as brute force attack. Hence, to solve these problems, a more powerful encryption technique has the following requirements for data confidentiality: robustness against cryptanalysis by providing extremely high complexity and capability to correct bit errors in low-SNR environments.

However, it is not efficient to encrypt short bit blocks because AES requires padding bits to fill out its 128-bit block size. When we introduce the cipher feedback (CFB) mode, it is possible to convert a block cipher into a stream cipher. The CFB mode can select an encryption block size (EBS) as desired in real-time cooperation with AES [12]. Because the previous ciphertext is used for encryption, attackers cannot know the data without the previous ciphertext even if they know the key. The streaming mode solves the problem of synchronization error from the burst bit error by the avalanche effect.

Furthermore, channel coding, such as turbo codes and low-density parity-check (LDPC) codes, can be utilized to satisfy the second requirement of bit error correction at low SNR. Turbo codes [15] encode data with low complexity at the satellite and decode with high complexity at the receiver, while LDPC codes perform in the opposite manner, which makes Turbo codes more appropriate for power limited satellite communications. Moreover, they achieve very low bit error rates (BERs) and converge to the Shannon limit in a low-SNR environment with a small number of iterations and low code rates [16]. Turbo codes have been recently adapted to 4G long-term evolution (LTE) [17], and are widely used in many communication systems, such as digital video broadcasting–return channel via satellite (DVB–RCS), DVB–RCS2, and NASA missions (Mars reconnaissance orbiter). A turbo encoder consists of two recursive systematic convolutional encoders, which are divided by an interleaver to form parallel concatenation. Turbo codes have the default code rate of 1/3, and it is possible to control the rate by puncturing. A turbo decoder comprises serial concatenation of two maximum a posteriori probability decoders that use the log-likelihood ratio for the iterative DEC algorithm. After a certain number of iterations, the decoder calculates the iterative DEC value. A large number of iterations can achieve low BERs.

Encryption key exchange as the encryption mechanism between consensus parties is another attractive issue in network security of satellites, which periodically orbit Earth at high speed. Boneh and Franklin [18] developed a method called identity-based encryption (IBE), which uses the identity (ID) information of the receiver, such as serial number, location, and travel route, as a public key. A third party, a public key generator (PKG), generates a private key by mean of the receiver ID and sends it back to the receiver; thus, the receiver does not need each of public and private key pairs of all the satellites. Multiple transmitters and one receiver can exchange symmetric encryption keys using these single key pairs. Through Weil pairing, a symmetric encryption key is encrypted with high security based on the bilinear Diffie–Hellman (BDH) assumption [19].

In this article, we suggest a novel method of joint encryption and channel coding called the CFB-AES-TURBO method, which encrypts data in the upper layer and conducts channel coding in the physical layer as bulk encryption with the key exchange method for satellite communication networks. The encryption key is negotiated through satellite and terrestrial security links, and the involved parties communicate the encrypted data through satellite links (Fig. 2). The CFB mode with the chaining property that enhances security performance can convert AES to stream cipher with the desired bit size selection (BS). Here, we select a bit size determined by trade-off between throughput, channel condition, and the security level required for a user, which has not been attempted thus far. Additionally, a use of turbo codes show better error correction at low-SNR environments due to the long propagation delay, weather attenuation, burst bit errors, and limited power in satellite communications. Therefore, the avalanche effect of AES can be solved by the CFB mode with EBS selection, and the BER performance can drastically improve with turbo codes. We derive the processing time gain and analyze

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computational complexity. We show the BER performance of the received and decrypted data with respect to the EBS and SNR. The security performances on brute-force and meet-in-the-middle attacks and indistinguishability are evaluated, which implies that satellite channel conditions should be considered for security. Finally, we drive the total expected latency of the proposed method. The contributions of this article can be summarized as follows:

1) We present the encryption/encoding (ENC) process and DEC/decryption process algorithms of CFB-AES-TURBO with EBS selection and a key exchange procedure.
2) We provide performance analysis, including complexity, BER performance, security performance, and total expected latency, by a use of the CFB mode with EBS selection and turbo codes.
3) We emphasize a need of considering satellite channels for evaluating security performance, and define the minimum key refresh time of GEO satellites to promote security in terms of latency.

The rest of this article is organized as follows. Section II presents related work on data confidentiality and key exchange. In Section III, we described the encryption/ENC and DEC/decryption processes of the proposed CFB-AES-TURBO method based on the control of EBS as well as the key exchange method between satellite, receivers, and PKG. Section IV presents the performance analysis of the proposed CFB-AES-TURBO method in terms of communication, security, and latency. Finally, Section V concludes this article.

II. RELATED WORK

For data confidentiality and key management, many investigations about superencryption and key exchange methods have been published. Encryption with error correction codes was first proposed by McEliece [20]. It was a public-key cryptosystem based on algebraic codes. However, there is a drawback that a large block size is required for error correction. Rao and Nam [21] proposed another public-key cryptosystem using a short block size to resolve the problem. Ning *et al.* [22] proposed AES and LDPC coding with easy key management that enhances security using a short encryption key. Pisek *et al.* [23] proposed joint security and advanced LDPC-based encryption and dual-step secure LDPC codes for channel coding that use AES and quasicyclic-LDPC (QC-LDPC) coding. A joint encryption, channel coding, and modulation scheme in [24] is based on the QC-LDPC lattice-code. The authors proposed encryption and channel coding using QC-LDPC codes and modulation using the coded modulation scheme of the lattice-code in a single step. With the erroneous channel characteristic of wiretap, the channel ENC of ciphertext produced by a bitwise XOR operation with plaintext, and linear feedback shift register (SR) sequence [25] achieved high-security performance for correlation attacks from Eve [26]. Errors in the wireless channel are of two-step ciphering. After DEC with a higher error probability of the wiretap channel than the main channel, Eve always has a higher error rate than the main receiver.

Hakan *et al.* [27] inserted the turbo coding process into the AES algorithm. They conducted turbo ENC and DEC after substitute byte transformation in the first round of encryption, and inverse shift row transformation in the last round of decryption. Studies to ensure security by adjusting code rates also have been conducted. Payandeh *et al.* [28] proposed a joint security and reliability method in one process through punctured turbo coding. They used a key to create a puncturing matrix with no encryption method, determined the code rate based on the channel state information of the channel estimator, and then produced the puncturing matrix with regard to the determined code rate using the pseudorandom number of the shared secret key. Mao and Qin [29] generated a puncturing matrix P with a shared secret key. Their method selects a 1/3 code rate when the element of $P(i)$ is 1, but it is increased to 1/2 while transmitting the information and parity bits of the first (second) encoder when i is odd (even) in case of $P(i) = 0$. Nguyen and Barbot [30] increased the 1/2 code rate by transmitting information and parity bits of the first (second) encoder when $P(i) = 0\,(P(i) = 1)$.

There have been many publications on encryption key exchange that use the abovementioned encryption method for confidentiality. The elliptic curve cryptosystem is used for encryption key distribution [31], but it requires an enormous number of pairs of public and private keys for all of the satellites in the network. For example, 66 key pairs are necessary in the Iridium satellite constellation, which degrades the efficiency. Efficient key distribution methods based on IBE were suggested in [32] and [33]. These methods address the disadvantage of requiring many key pairs because they use ID for a public key, which enhances the cost-efficiency of key exchange and management. For a space–terrestrial integrated network, an emergency security scheme was proposed [9]. It is composed of emergency data transmission, group block design-based key agreement, and secure satellite information acquisition. Each phase uses its own secret key and another key from a key generation center with hash functions and bilinear pairing.

We deal with both confidentiality and key exchange among the important requirements, whereas previous studies have taken into account either confidentiality or key exchange separately. We introduce the CFB mode, whose bitwise XOR operation is substituted for the nonlinear mapping of conventional communication systems. Thus, we address burst bit errors after AES decryption and enable real-time operation. We propose the addition of turbo codes at the end of each encryption block, which drastically reduces both the BER and processing time. To improve the resource efficiency of data transmission, we make an effort to control the EBS. We take the key exchange into account based on the staged IBE (SIBE) before data encryption.

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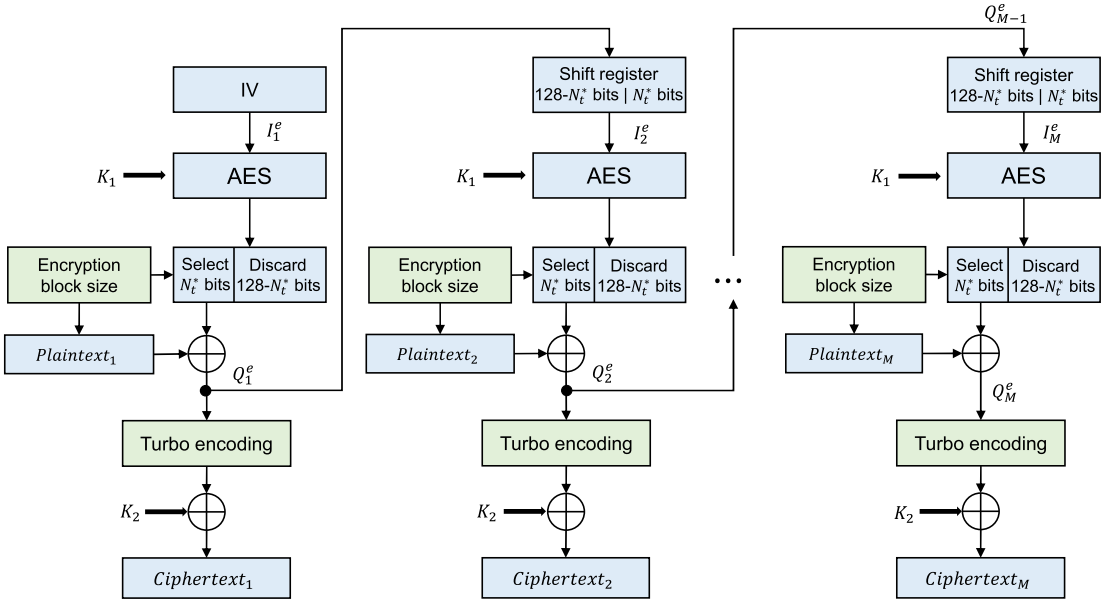


Fig. 3. Encryption and ENC process of the proposed CFB-AES-TURBO method.

III. PROPOSED JOINT SECURITY METHOD

In this section, we describe our proposed CFB-AES-TURBO method, which integrates encryption and channel coding with key exchange. First, we present the processes of CFB-AES-TURBO, and then explain how the EBS is controlled based on channel conditions. We present the detailed encryption key pair exchange based on the SIBE in sequence.

A. Encryption and Encoding

Fig. 3 shows the encryption/ENC process. To control the EBS according to channel conditions, we introduce the CFB mode, in which a block cipher is converted into a stream cipher. The mth ciphertext, ciphertext_m, is derived by

$$I_m^e = \text{LSB}_{128-N_t^*}(I_{m-1}^e)||Q_{m-1}^e \quad (1)$$

$$Q_m^e = \text{plaintext}_m \oplus \text{MSB}_{N_t^*}(E[K_1, I_m^e]) \quad (2)$$

$$\text{ciphertext}_m = \text{turbo_encoding}(Q_m^e) \oplus K_2 \quad (3)$$

where I_m^e is the mth SR ($||$) of the encryption/En process and Q_m^e is the result of the mth XOR operation ($\oplus$). Here, N_t^* is the desired EBS, $\text{LSB}_{128-N_t^*}(\cdot)$ represents the tail $128 - N_t^*$ bits (least significant) of the SR, and $\text{MSB}_{N_t^*}(\cdot)$ represents the head N_t^* bits (most significant) of the AES encryption result. Further, K_1 and K_2 represent the encryption key in AES encryption and the key for the turbo codes, respectively. The result of encryption $E[K_1, \cdot]$ uses the encryption key K_1, and the size of the mth segmented plaintext, plaintext_m, is equal to N_t^*.

Key exchange (with details in Section III-D) can be performed for a key pair (K_1, K_2) between a satellite and receivers (e.g., gateway, base station, another satellite, aircraft, etc.) with parameters from PKG. AES is operated with the initialization value (IV) and K_1. Then, we calculate an EBS of N_t^* bits, select $\text{MSB}_{N_t^*}(E[K_1, I_m^e])$, and discard $\text{LSB}_{128-N_t^*}(E[K_1, I_m^e])$. The plaintext must be segmented into N_t^* bits for encryption. Equation (2) states the XOR operation with $\text{MSB}_{N_t^*}(E[K_1, I_m^e])$ and the segmented plaintext. After turbo encoding in (3), XOR operation is conducted again with K_2. In parallel, Q_m^e moves to the next block, and then settles behind the tail of the $(m+1)$th SR. We repeat these steps and then obtain Q_{m+1}^e, simultaneously. In a similar way with [24], the CFB-AES-TURBO can be extended to a joint modulation scheme in a single step if we use turbo trellis codes as coded modulation.

B. DEC and Decryption

DEC/decryption is the reverse of encryption/ENC, as shown in Fig. 4. The plaintext_m is expressed by

$$I_m^d = \text{LSB}_{128-N_t^*}(I_{m-1}^d)||Q_{m-1}^d \quad (4)$$

$$Q_m^d = \text{turbo_decoding}(\text{ciphertext}_m) \oplus K_2 \quad (5)$$

$$\text{plaintext}_m = Q_m^d \oplus \text{MSB}_{N_t^*}(E[K_1, I_m^d]) \quad (6)$$

where I_m^d is the mth SR of the decryption/DEC process and Q_m^d is the result of the mth turbo DEC.

The receiver can obtain the encryption key pair (K_1, K_2) through the process of key exchange. It then calculates (4) and (5) in parallel to derive $\text{MSB}_{N_t^*}(E[K_1, I_m^d])$. Then, the plaintext is derived by the XOR operation expressed in (6). At the same time, the value of Q_m^d in (5) is sent to the $(m+1)$th tail of the SR in the next block, where the next AES encryption is carried out. In the DEC/decryption process, the mth AES and the mth turbo DEC can be activated concurrently. Here, we note the two following points. First, (2) is identical with (5), as (1) is with (4). These can be easily shown by

$$\begin{aligned} Q_m^e &= \text{turbo_encoding}^{-1}(\text{ciphertext}_m) \\ &= \text{turbo_decoding}(\text{ciphertext}_m) \\ &= Q_m^d \end{aligned} \quad (7)$$

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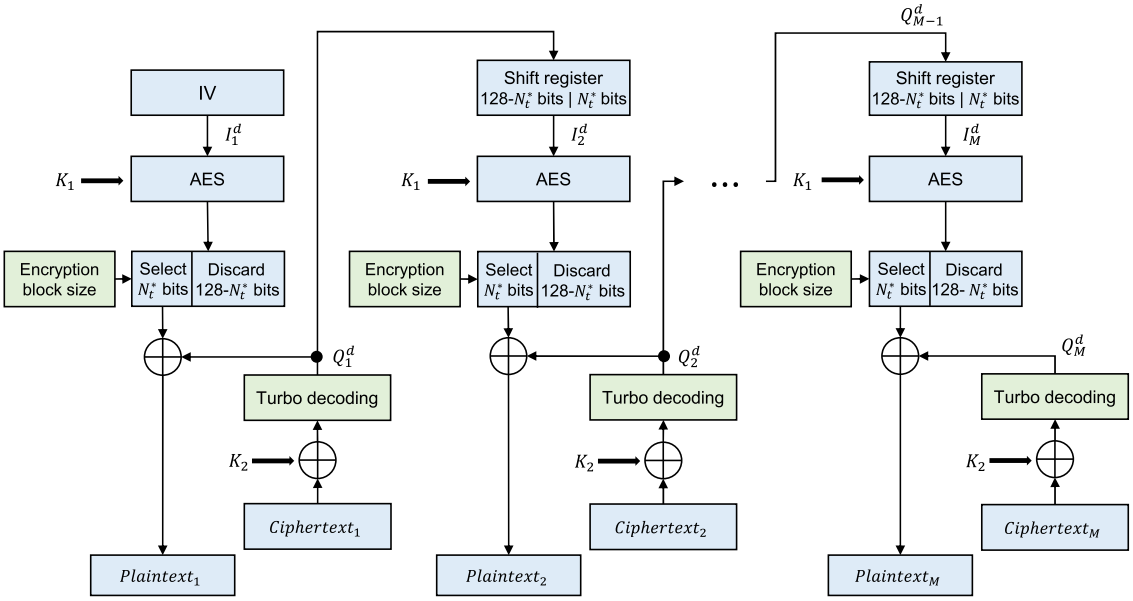


Fig. 4. DEC and decryption process of the proposed CFB-AES-TURBO method.

where the inverse process of turbo ENC turbo_encoding^{-1} is exactly equivalent to turbo_encoding, and

$$\begin{aligned} I_m^e &= \mathrm{LSB}_{128-N_t^*}(\mathrm{IV})||Q_{m-1}^e \\ &= \mathrm{LSB}_{128-N_t^*}(\mathrm{IV})||Q_{m-1}^d \\ &= I_m^d. \end{aligned} \tag{8}$$

Second, the proposed CFB-AES-TURBO method does not require the AES decryption algorithm because of the property of the CFB mode. This can be shown by

$$Q_m^e = \mathrm{plaintext}_m \oplus \mathrm{MSB}_{N_t^*}(E[K_1, I_m^e]) \tag{9}$$

$$\begin{aligned} \mathrm{plaintext}_m &= Q_m^e \oplus \mathrm{MSB}_{N_t^*}(E[K_1, I_m^e]) \\ &= Q_m^d \oplus \mathrm{MSB}_{N_t^*}(E[K_1, I_m^d]) \end{aligned} \tag{10}$$

where the values of $\mathrm{MSB}_{N_t^*}(E[K_1, I_m^e])$ and $\mathrm{MSB}_{N_t^*}(E[K_1, I_m^d])$ are identical to obtain the right plaintext. Thus, we use only the AES encryption algorithm. Since we adopt the CFB mode and add turbo codes at the end of each block, we can execute the joint security method block-by-block in real time, which can adjust the data length, modify the bit errors from wireless channels, and reduce the synchronization error at the same time.

C. Encryption Block Size Selection

As shown in Fig. 1, AES has a nonlinear function in the form of S-box, which can result in the avalanche effect. Bit errors in the wireless environment may cause the throughput to decrease. Thus, to determine the optimal EBS of N_t^*, our problem to maximize EBS N_i under the required security level S_{req}, and the maximum bit size $N_{\max}$ constraints is given by

$$\max \frac{1}{WR_{\max}} \sum_{i=t-W+1}^{t} R_i(1-N_iP_i) \tag{11}$$

$$\text{s.t.} \frac{1}{WS_{\max}} \sum_{i=t-W+1}^{t} \log_2 N_i = S_{\mathrm{req}} \tag{12}$$

$$N_i \le N_{\max}. \tag{13}$$

The throughput $R_i(1-P_i)^{N_i} \approx R_i(1-N_iP_i)$ is normalized by $R_{\max} = \max\{R_i\}$, where R_i and P_i represent the transmission rate and the uncoded BER, respectively. The required security level (12) is defined by EBS normalized by $S_{\max} = \log_2 N_{\max}$ [14]. Since it is difficult to perfectly know the channel condition using the feedback system from ground due to long latency of satellite communications, we consider a causal system by means of window size W. Further, the maximum EBS (128 bits in the AES) is taken into account in (13). Through the Karush–Kuhn–Tucker condition [34], the optimal EBS at transmission time t is derived by

$$N_t^* = \min\left\{N_{\max}, \frac{(\prod_{i=t-W+1}^{t} P_i)^{\frac{1}{W}}}{P_t} e^{S_{\max}S_{\mathrm{req}}\ln 2}\right\} \tag{14}$$

which gives a larger block size when P_t is smaller. This allows users in a low-BER environment (i.e., better channel condition) to encrypt more data. Here, original data is split into M pieces for the encryption and ENC process as explained in Section III-A, and then transmitted for M consecutive time slots. Since the processing time is much shorter than the coherence time of slow-fading satellite channel attenuation, M time slots are in the quasistatic channel condition, which results in the same encryption size N_t^* for all M pieces.

In practice, long latency and up/downlink channel difference in frequency-division duplexing should be taken into account for satellite channel prediction [35] and BER estimation [36]. To transmit many frames consisting of multiple encryption blocks simultaneously, we can use frame

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aggregation and then achieve high throughput, as currently deployed in the wireless local area network [37]. We note that the BER lowered by turbo coding influences the optimal EBS. For example, we can control the number of turbo DEC iterations to determine the desirable EBS, which is beyond the scope of this article.

D. Key Exchange

To exchange the encryption key pair (K_1, K_2) exploited in the CFB-AES-TURBO method, we adopt the SIBE [33]. Since a satellite moves around the Earth in a certain orbit with high speed and both satellite and receiver should agree on key pair every encountering time, the SIBE, which periodically shares confidential data with the control center, is suitable for satellite communication networks. The encryption key pair is encrypted by means of a public key of the receiver, which has the ID form, and it is decrypted by a private key made from the PKG. This is made up in six steps. We assume that receivers broadcast their IDs to satellites, periodically.

1) `Setup`: The PKG generates two groups $\mathbb{G}_1$ and $\mathbb{G}_2$ based on prime order q, and creates an admissible bilinear map, $e = \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_2$, using the two groups. The PKG picks s as its private key from the set Z_q of modulo operation of q and calculates $y = g^s$, where g is the generator of the two groups. The PKG also selects three hash functions H_1, H_2, and H_3 among SHA-256, SHA-384, SHA-512, and RIPEMD-160 [38]. The hash function with arbitrary string $*$, $H_1 : \{0, 1\}^* \rightarrow \mathbb{G}_1$, is used to create K_{ID}, which signifies a private key of the receiver in the `Extract`. Other hash functions, $H_2 : \{0, 1\}^n$ and $H_3 : \{0, 1\}^{N_t^* \times 3}$, take advantage of the `KEYEnc`. Finally, the PKG transmits public parameters $= \{\mathbb{G}_1, \mathbb{G}_2, e, g, y, H_1, H_2, H_3\}$ to both the satellite and the receiver.
2) `Extract`: The PKG that has received the receiver ID computes $K_{\mathrm{ID}} = H_1(\mathrm{ID})^s \in \mathbb{G}_1$ as a private key and sends it to the receiver.
3) `KEYEnc`: The satellite in the management area of the receiver receives the ID and public parameters from the receiver and the PKG, respectively. The satellite creates AES encryption key K_1 and turbo codes key K_2 in the key space $\{0, 1\}^n$ and $\{0, 1\}^{N_t^* \times 3}$, and picks $r \in Z_q$ randomly. The encrypted key pair, C_K, is calculated by

$$\mu = g^r \tag{15}$$
$$\nu = H_2(e(H_1(\mathrm{ID}), y)^r) \oplus K_1 \tag{16}$$
$$\omega = H_3(e(H_1(\mathrm{ID}), y)^r) \oplus K_2 \tag{17}$$
$$C_K = \mathrm{turbo_encoding}(\mu, \nu, \omega). \tag{18}$$

4) `DATAEnc`: The satellite with the key pair sends the encrypted data C_D based on the procedure in Section III-A.

TABLE I
Parameters After Each Key Exchange Process

Step	PKG	Receiver	Satellite
Precondition	ID	ID	ID
`Setup`	Public parameters $S(private\ key)$	Public parameters	Public parameters
`Extract`	K_{ID}	K_{ID}	
`KEYEnc`			K_1(AES) K_2(Turbo) C_K
`DATAEnc`			C_D
`KEYDec`		K_1(AES) K_2(Turbo)	
`DATADec`		$data$	

5) `KEYDec`: The receiver that has received C_K from the satellite can obtain the key pair as follows:

$$(\mu, \nu, \omega) = \mathrm{turbo_decoding}(C_K) \tag{19}$$
$$K_1 = \nu \oplus H_2(e(K_{\mathrm{ID}}, \mu)) \tag{20}$$
$$K_2 = \omega \oplus H_3(e(K_{\mathrm{ID}}, \mu)). \tag{21}$$

We note that $e(K_{\mathrm{ID}}, \mu) = e(H_1(\mathrm{ID}), y)^r$ by admissible bilinear pairing.[1]
6) `DATADec`: The receiver with the key pair decrypts the original data based on the procedure in Section III-B.

Table I gives core parameters. It is important that satellites and receivers communicate with `DATAEnc` and `DATADec` after the key exchange process, while the key exchange begins as soon as satellites enter the management area of receivers.

IV. PERFORMANCE ANALYSIS

In this section, we first derive the processing time and the gain of the proposed CFB-AES-TURBO method. We also show that the BER performance of the proposed method improves markedly in comparison to those of other methods. In addition, we analyze the enhanced security performance against two types of attacks and provide indistinguishability according to channel conditions. Then, we provide latency analysis.

A. Processing Time and Complexity

As seen in Figs. 3 and 4, the mth turbo ENC and the $(m + 1)$th AES encryption in the encryption/ENC process, and the mth turbo DEC and the mth AES encryption[2] in the DEC/decryption process are activated in parallel. Therefore, the processing time is faster than that of conventional communication systems that conduct channel coding after data encryption. The processing times of AES and turbo ENC/DEC depend on the central processing unit and

[1]The BDH problem holds the relationship of $e(A^a, B^b) = e(A, B)^{ab} = e(A^b, B^a)$, but it is infeasible to find A^b and B^a using only A^a and B^b [18].
[2]The AES decryption algorithm is not needed as given in (9) and (10).

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TABLE II
Processing Time and Gains of the Proposed Method

	Case	Gain
Encryption and encoding	(a)	$(M-1)T_{enc}$
	(b)	$(M-1)T_{aes}$
Decoding and decryption	(a)	MT_{dec}
	(b)	MT_{aes}

input size. Because it is difficult to compare the processing times of the two procedures, we consider two cases as follows: (a) $T_{\text{aes}} > T_{\text{enc}}$ and $T_{\text{aes}} > T_{\text{dec}}$ and (b) $T_{\text{aes}} < T_{\text{enc}}$ and $T_{\text{aes}} < T_{\text{dec}}$, where T_{aes}, T_{enc}, and T_{dec} denote the AES encryption time, the turbo ENC time, and the turbo DEC time, respectively.

The processing time of encryption/ENC for case (a), $T^e_{\text{cat(a)}}$, and case (b), $T^e_{\text{cat(b)}}$, are

$$\begin{aligned} T^e_{\text{cat(a)}} &= T_{\text{ebs}}+T_{\text{bs}}+T_{\text{ps}}+T_{\text{sr}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}}+T_{\text{enc}} \\ &\approx T_{\text{ebs}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}}+T_{\text{enc}} \qquad (22) \\ T^e_{\text{cat(b)}} &= T_{\text{ebs}}+T_{\text{bs}}+T_{\text{ps}}+T_{\text{sr}}+T_{\text{aes}}+4MN_t^*T_{\text{xor}}+MT_{\text{enc}} \\ &\approx T_{\text{ebs}}+T_{\text{aes}}+4MN_t^*T_{\text{xor}}+MT_{\text{enc}} \qquad (23) \end{aligned}$$

where T_{ebs} and T_{xor} are the calculation time for the EBS of N_t^* bits and XOR operation time for one bit, respectively. Here, T_{bs}, T_{ps}, and T_{sr} represent the times of desired BS, plaintext segmentation (PS), and settling at the end of SR, respectively. It is reasonable to suppress those operation times because they consume shorter time only for cutting and moving relative to the other processing times. The processing time of DEC/decryption for case (a), $T^d_{\text{cat(a)}}$, and case (b), $T^d_{\text{cat(b)}}$, are

$$\begin{aligned} T^d_{\text{cat(a)}} &= T_{\text{ebs}}+T_{\text{bs}}+T_{\text{sr}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}} \\ &\approx T_{\text{ebs}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}} \qquad (24) \\ T^d_{\text{cat(b)}} &= T_{\text{ebs}}+T_{\text{bs}}+T_{\text{sr}}+4MN_t^*T_{\text{xor}}+MT_{\text{dec}} \\ &\approx T_{\text{ebs}}+4MN_t^*T_{\text{xor}}+MT_{\text{dec}}. \qquad (25) \end{aligned}$$

On the other hand, the conventional communication system, which executes channel coding after encryption at the upper layer individually have the processing time of encryption/ENC, T^e_{con}, and DEC/decryption, T^d_{con}, given as

$$\begin{aligned} T^e_{\text{con}} &= T_{\text{ebs}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}}+MT_{\text{enc}} \qquad (26) \\ T^d_{\text{con}} &= T_{\text{ebs}}+MT_{\text{aes}}+4MN_t^*T_{\text{xor}}+MT_{\text{dec}}. \qquad (27) \end{aligned}$$

Since encryption and channel coding are separated, there is no difference between the two cases, (a) and (b), in the conventional communication system. The processing time gains derived by calculating $T^e_{\text{con}} - T^e_{\text{cat}}$ and $T^d_{\text{con}} - T^d_{\text{cat}}$ for each case, as given in Table II. The gains depend on T_{enc}, T_{dec}, and T_{aes} in both cases (a) and (b) because the AES and turbo codes are concurrently activated on each block. Therefore, we can achieve a much greater processing time gain when we transmit large size data.

In the CFB-AES-TURBO method, there are eight parts of operations in encryption/ENC, including EBS, AES, BS, PS equal to BS, ENC, SR, and two XOR operations. In DEC/decryption with DEC instead of ENC, there are seven parts of operations, except for PS because the received ciphertext is decoded and decrypted immediately without segmentation of ciphertext. EBS is conducted at the first time on the first block before PS, which gives the constant complexity of $\mathcal{O}(1)$. Also, AES, BS, PS, and SR with constant complexity for every block give $\mathcal{O}(M)$ complexities. Besides, turbo ENC/DEC and XOR operations depending on input block size for every block give $\mathcal{O}(MN_t^*)$ complexities [39]. Therefore, the total computational complexities of encryption/ENC and DEC/decryption in the CFB-AES-TURBO are given by

$$\mathcal{O}(3MN_t^*+4M+1)=\mathcal{O}(MN_t^*) \qquad (28)$$

$$\mathcal{O}(3MN_t^*+3M+1)=\mathcal{O}(MN_t^*) \qquad (29)$$

which has dependence on data size totally.

B. BER Performance

In Section I, we noted the importance of the received data quality. Under space radiation environments, a single event upset (SEU) in the onboard processor (OBP) satellite is inevitable, which induces a malfunction on the proposed security method. A system failure rate can be estimated according to the SEU rate, the number of mitigation windows, and OBP shield thickness [40]. The system errors by the radiation effect can be addressed by means of the radiation effects and analysis group external blind scrubber. Moreover, error propagations can be prevented to all subsequent blocks by the self-recovering property [41].

We now provide numerical results to evaluate the BER performance of our CFB-AES-TURBO method by adjusting the EBS according to channel condition. Fig. 5 shows the BER curves in the Rician fading satellite channel with the K-factor of 10, where we use binary phase shift keying modulation. In order to illustrate the BER performance with respect to the channel condition and EBS, Fig. 5 uses fixed EBSs of 40, 72, 96, and 128 bits for turbo codes with the code rate of 1/3, specified in 4G LTE [17], and channel conditions from 0 to 4 dB of SNR[3]. As we increase the EBS from Fig. 5(a) to (d), lower BER performance is achieved, benefiting from large interleaver size. For example, with 72 bits of the encryption block, we can obtain a BER lower than 10^{-5}at the SNR of 4 dB with a coding gain of about 12 dB[4] as the same BER is achieved in the Rician fading channel at the SNR of about 16 dB [42, Fig. 4.12].

When we consider instantaneous channel condition (i.e., $W = 1$), the optimization problem in (11) $\sim$ (13) is simplified as

$$\max R_t(1-P_t)^{N_t} \qquad (30)$$

[3]The BER performances with various code rates based on the channel conditions are reported in [1], but since even one bit error should not be allowed to happen to the encryption key during key exchange process, we do not control the code rate in this work.

[4]In the additive white Gaussian noise channel, a coding gain of about 5 dB can be obtained [1].

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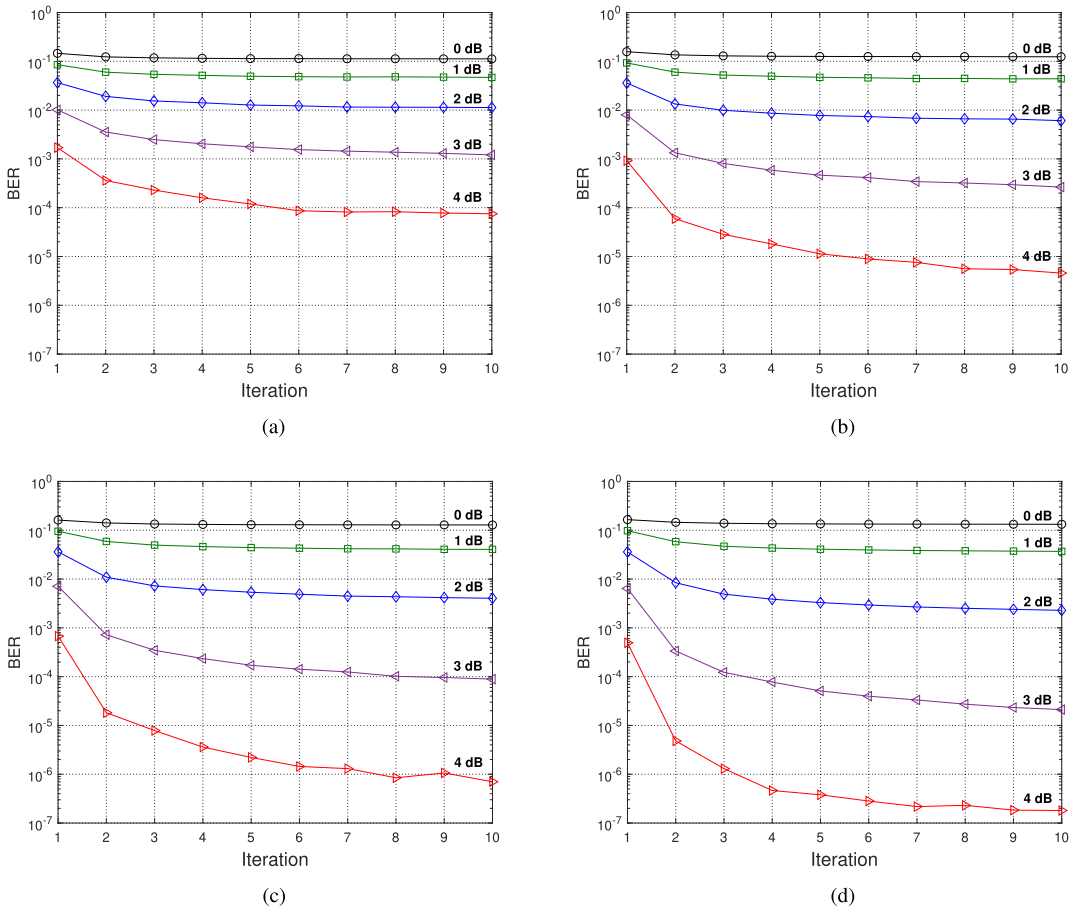


Fig. 5. BER performance with respect to the number of iterations according to channel conditions with considering window size at EBS of (a) 40 bits, (b) 72 bits, (c) 96 bits, and (d) 128 bits.

$$\text{s.t.}\frac{1}{S_{\max}}\log_2 N_i = S_{\text{req}} \tag{31}$$

$$N_t \le N_{\max} \tag{32}$$

which gives the optimal EBS N_t^* as follows:

$$N_t^* = \min\left\{N_{\max}\right.$$
$$\left.\frac{1}{2}\left[\frac{1}{P_t}+\sqrt{\left(\frac{1}{P_t}\right)^2+2^{S_{\max}S_{\text{req}}+2}\left(2^{S_{\max}S_{\text{req}}}-\frac{1}{P_t}\right)}\right]\right\}. \tag{33}$$

Fig. 6 shows the BER performance in the same environment as in Fig. 5. For every channel condition, the optimized encryption block of the simplified problem has 64 bits[5] by (33), which leads to worse BER performances than those of Figs. 5(b), (c), and (d) due to a small interleaver size. In practice, the window size W plays an important role to determine the EBS, which raises a new future problem of choosing the optimal window size for better performances.

[5]In fact, the optimized encryption block of the simplified problem may have the decimal number size. However, because the interleaver of turbo codes supports a size of the natural number [43], we use a floor function $\lfloor\cdot\rfloor$.

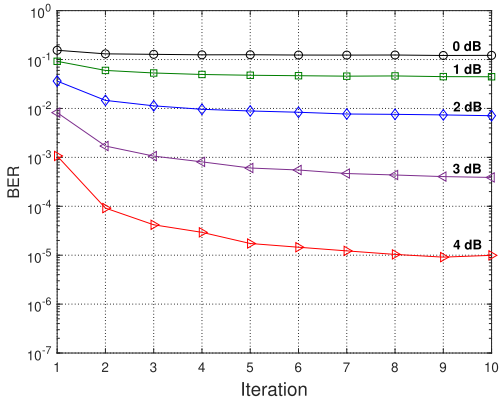


Fig. 6. BER performance with respect to the number of iterations according to channel conditions with window size $W = 1$.

Despite a few literatures on joint block cipher and turbo codes, it is difficult to draw meaningful results for satellite communication systems. With another scheme of block cipher with QC-LDPC [24], a good symbol error rate performance is achieved at high-SNR environment, which is not a typical satellite link condition with low SNR. With an assumption of perfect DEC in previous blocks for the

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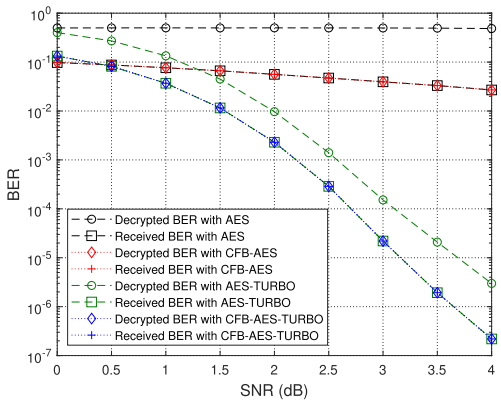


Fig. 7. BER comparison with $N_t^* = 128$ for ten iterative DEC in Rician fading channel with K-factor equal to 10.

received and decrypted data, Fig. 7 shows BER performance comparison between the proposed method in case of the maximum 128 bits in Fig. 5(d), and the traditional encryption methods as benchmarks. Even if errors are not corrected after turbo DEC, error propagations can be prevented by self-recovering. The received data denotes erroneous data by the satellite channel before decryption, while the decrypted data denotes the plaintext after decryption. For AES only, the BER of the decrypted data is much higher than that of the received data because AES decryption has inverse substitute byte transformation that uses the reverse S-box. The received BER performance of the AES achieves exactly the same as that of both the received and decrypted CFB-AES due to the bitwise XOR operation of the CFB mode, and the AES-TURBO and CFB-AES-TURBO also achieve the same performance as well. In the AES-TURBO, which performs channel coding after AES, the BER of the received data is significantly low, but the BER of the decrypted data is much higher than that of the received data with traditional AES. On the contrary, the proposed CFB-AES-TURBO method with 10 iterative turbo DEC shows markedly low BER, thanks to the CFB mode and turbo codes, achieving a BER of about 10^{-7} for both the received and decrypted data at an SNR of 4 dB with the maximum EBS. Especially, the use of CFB-AES-TURBO leads to better BER performances than AES, CFB-AES, and AES-TURBO after decryption at 4-dB SNR by 0.4850, 0.0269, and 2.7301×10^{-7}, respectively.

Due to additional functionalities of CFB mode and turbo codes, it is inevitable to incur additional cost in a power-limited satellite system. Therefore, we are faced with a trade-off problem between BER performance and resource utilization. Efficient control of the EBS also leads to additional operation cost. However, it is manageable because large-scale fading, which is dominant in satellite communications, does not induce drastically changing channel attenuation in the order of seconds [35]. The proposed method, thus, does not need to calculate parameters for every transmission.

TABLE III
Security Performance

	Case 1	Case 2
Brute-force attack	$2^n \times 2^{N_t^* \times 3}$	$\prod_{m=0}^{M-1}(2^n - m)(2^{N_t^* \times 3} - m)$
Meet-in-the-middle attack	$2^n + 2^{N_t^* \times 3}$	Unavailable

C. Security Performance

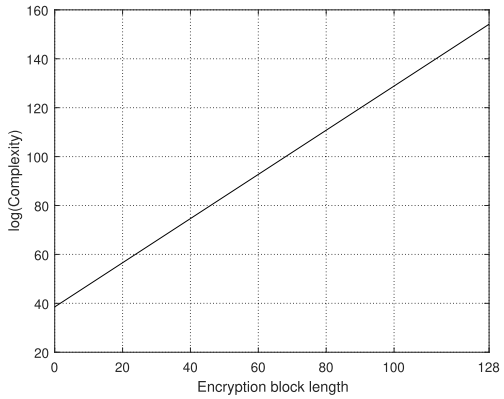
Here, we analyze the security performance of the proposed CFB-AES-TURBO method by evaluating the resistances against the brute-force and meet-in-the-middle attacks, which are basic cryptanalysis and cryptanalysis of the two-step encryption system, respectively.

We divide the analysis into the two cases for the attacks. Case 1 uses the same key pair (K_1, K_2) for every block aforementioned, and Case 2 uses a different key pair (K_{1m}, K_{2m}) for the mth block [14]. If we use a bitstream generator, random encryption keys can be generated in satellites and receivers with no synchronization error. Table III gives the security performance for the two attacks.

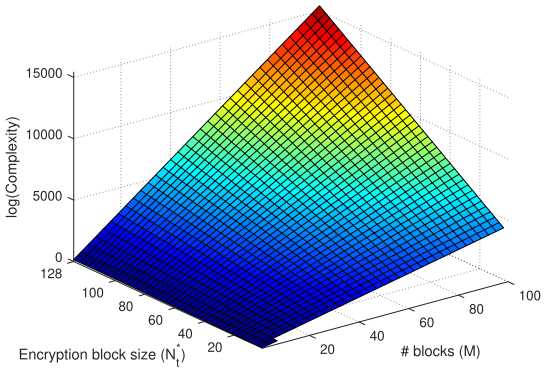
For Case 1, the attacker needs to know K_1 with 2^n complexity, where n is one of the feasible key types (128, 192, or 256 bits). In addition, K_2 should be obtained to derive the key of $2^{N_t^* \times 3}$ complexity. Hence, it results in the total complexity of $2^n \times 2^{N_t^* \times 3}$ for the brute-force attack. Meanwhile, our proposed CFB-AES-TURBO method is superencryption in the two steps, because it requires two keys for encryption and ENC. This implies that the proposed CFB-AES-TURBO method is exposed to the meet-in-the-middle attack. Attackers calculate complexity and make two tables that contain the results of all the possible 2^n encryption keys for AES encryption, and all the possible $2^{N_t^* \times 3}$ keys for turbo DEC. Attackers find the key pairs that have the equal sequence of the abovementioned two tables, and then conduct α encryptions for $(\text{plaintext}_2, \text{ciphertext}_2)$ to obtain the correct key pair. Therefore, the total complexity is $2^n + 2^{N_t^* \times 3} + \alpha \approx 2^n + 2^{N_t^* \times 3}$ for the meet-in-the-middle attack, which still provides higher security performance than using no second-step encryption.

For Case 2, under the brute-force attack, the encryption key requires the complexity of $2^n \times (2^n - 1) \times \cdots \times \{2^n - (M-1)\}$, where M is the number of blocks, and the K_2 results in the computation of $2^{N_t^* \times 3} \times (2^{N_t^* \times 3} - 1) \times \cdots \times \{2^{N_t^* \times 3} - (M-1)\}$. Thus, the total computational costs are the amount of $\prod_{m=0}^{M-1}(2^{N_t^* \times 3} - m)(2^n - m)$. Fig. 8 provides computational complexities in the range of N_t^* from 1 to 128 bits, as measuring security strength against brute-force attacks. For Case 1 in Fig. 8(a), it is not necessary to consider the value of M because it uses the same key pair for every block. On the other hand, Case 2 in Fig. 8(b) exploits a different key pair for each block and takes $M = 1, 2, \ldots, 100$ into account for performance evaluation. It demonstrates the extremely high complexity, at maximum about 10^{15000} times higher than using the same key pair in Fig. 8(a), as N_t^*

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(a)



(b)

Fig. 8. Complexity comparison between (a) same key pair in Case 1 and (b) different key pairs in Case 2 under brute-force attack with encryption key $n = 128$ bits.

and M increase. We note that an attacker cannot attempt the meet-in-the-middle attack because the data are encrypted with a different key pair for each block. Despite extremely high-security performance with different key pairs, it is unavoidable that key management of a large volume of memories with a large number of encryption key pairs should be a burden to satellites. The rainbow table [44] is one of the solutions to resolve key management.

Under a jamming attack, we remark that code division multiple access (CDMA) can protect the desired signal from the jamming signal using a specific pseudorandom code. Moreover, in a disguised jamming attack that mimics the desired signal by means of the identical chip rate with the conventional scrambling code, CDMA with AES as a secure scrambling can break the symmetricity between the desired signal and the disguised one, which increases the secrecy capacity significantly [45].

D. Indistinguishability

As an adversary pretends to be a legitimate receiver with the correct pair of plaintext and ciphertext, it can discover the encryption key using the relationship of the pair, which is called the chosen plaintext attack (CPA). To escape the threat of CPA, indistinguishability has to be provided, so that the attacker cannot distinguish which data

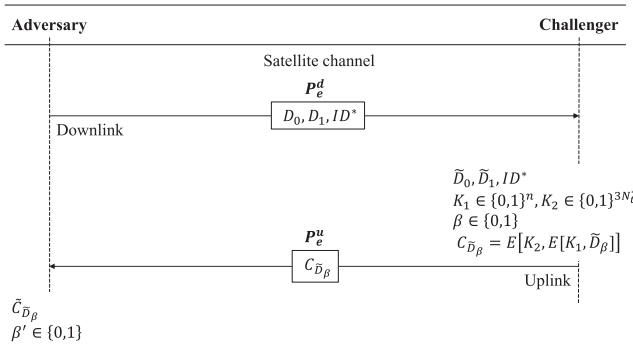


Fig. 9. Process of indistinguishability in the satellite channel.

was encrypted [18], [19]. Indistinguishability is considered a game between adversary $\mathcal{A}$ and challenger $\mathcal{C}$. $\mathcal{A}$ transmits two equal length data (D_0, D_1) to $\mathcal{C}$, which is considered the most difficult game, and then $\mathcal{C}$ sends the data after encrypting either D_0 or D_1 probabilistically. Upon receiving the encrypted data, $\mathcal{A}$ guesses which of the two messages was encrypted. The steps are described in detail as follows:

1) $\mathcal{A}$ issues the private key extraction queries about $\mathrm{ID}_1, \ldots, \mathrm{ID}_l$. $\mathcal{C}$ transmits K_{ID_a} of ID_a for the identity information of the ath query to $\mathcal{A}$ through the `Extract` algorithm.
2) $\mathcal{A}$ sends two equal length data (D_0, D_1) and target ID^* to $\mathcal{C}$.
3) $\mathcal{C}$ creates an encryption key pair (K_1, K_2), selects $\beta \in \{0, 1\}$, and then encrypts M_β. Sequentially, the encryption key pair encrypted under ID^* is passed to $\mathcal{A}$ with the ciphertext of D_β.
4) $\mathcal{A}$ can obtain the private key of ID_a with restriction of $\mathrm{ID}_a \neq \mathrm{ID}^*$.
5) $\mathcal{A}$ prints out a guess β' with respect to β.

$\mathcal{A}$ is the winner of this game with $\beta = \beta'$, leading to no indistinguishability. That is, the proposed method is exposed to CPA, and the advantage of $\mathcal{A}$ can be defined as $\mathrm{Adv}(\lambda) = |\mathbb{P}[\beta = \beta'] - 1/2|$, where λ is the security parameter [33]. We ensure that our proposed method is secure if the advantage is negligible, which is called semantically secure.

However, we need to consider burst bit errors from satellite communication links. In Fig. 9, $\tilde{D}_0$ and $\tilde{D}_1$ indicate incorrect messages from $\mathcal{A}$ due to satellite channel errors, and $C_{\tilde{D}_\beta}$ is the ciphertext of $\tilde{D}_0$ or $\tilde{D}_1$. We take into account the satellite channel conditions when $\mathcal{A}$ and $\mathcal{C}$ exchange the original and/or encrypted data. We should consider the following three points: a) $\mathbb{P}[\tilde{D}_\beta = D_{\beta'}]$, b) advantage $\mathrm{Adv}(\lambda)$, and c) $\mathbb{P}[C_{\tilde{D}_\beta} = \tilde{C}_{D_\beta}]$. Point a) is the probability that the data $\tilde{D}_\beta$, which is selected by $\mathcal{C}$ after being polluted by the satellite downlink channel from $\mathcal{A}$ and data $D_{\beta'}$, which is given by Step 5), are identical. The probability of $\mathbb{P}[C_{\tilde{D}_\beta} = \tilde{C}_{\tilde{D}_\beta}]$ in c) represents that the data $C_{\tilde{D}_\beta}$ to be encrypted by $\mathcal{C}$ and the data $\tilde{C}_{\tilde{D}_\beta}$ received from $\mathcal{C}$ after going through the satellite uplink channel are identical. Probabilities are represented by $\mathbb{P}[\text{no error}|\beta = \beta']$, with

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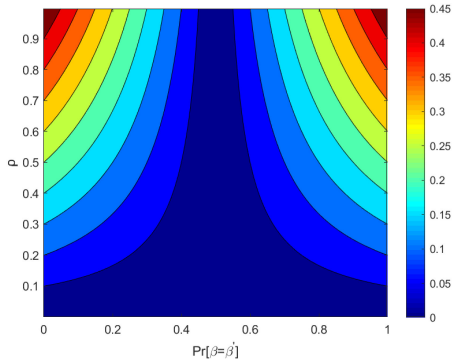


Fig. 10. Advantage according to the satellite channel qualities and $\mathbb{P}[\beta = \beta']$.

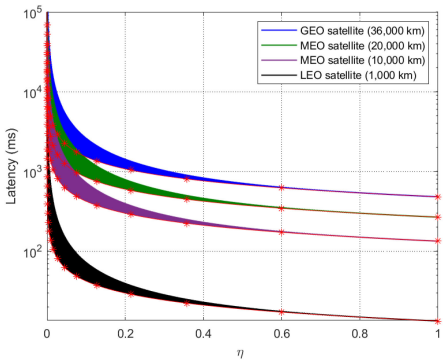


Fig. 11. Total expected latency of the CFB-AES-TURBO method according to altitude and η in $N_t^d = N_t^u = 128$ bits case.

which $\mathbb{P}[\text{no error}]$ and $\mathbb{P}[\beta = \beta']$ are independent of each other. Thus, we have

$$\mathbb{P}[\tilde{D}_\beta = D_{\beta'}] = (1 - P_e^d)^{N_t^*} \tag{34}$$

$$\mathbb{P}[C_{\tilde{D}_\beta} = \tilde{C}_{\tilde{D}_\beta}] = (1 - P_e^u)^{3N_t^*} \tag{35}$$

where N_t^* is the EBS, $3N_t^*$ is the transmitted data length, and P_e^d and P_e^u are the down/uplink coded BER, respectively, given in Section III-C and III-D. In (34) and (35), we assume that every bit is independent, and that burst bit errors are suppressed because information bits are scrambled sufficiently by the use of a scrambler and/or an interleaver.

The advantage is derived as follows:

$$\begin{aligned}\text{Adv}(\lambda) &= \mathbb{P}[\tilde{D}_\beta = D_{\beta'}] \times \left|\mathbb{P}[\beta = \beta'] - \frac{1}{2}\right| \\ &\quad \times \mathbb{P}[C_{\tilde{D}_\beta} = \tilde{C}_{\tilde{D}_\beta}] \\ &\approx \rho \cdot \left|\mathbb{P}[\beta = \beta'] - \frac{1}{2}\right| \end{aligned} \tag{36}$$

where $\rho = (1 - P_e^d)^{N_t^*} \times (1 - P_e^u)^{3N_t^*}$ represents the channel qualities with downlink BER P_e^d and uplink BER P_e^u obtained from Section IV-B for sufficiently large N_t^* and very small BERs. We can now degrade the advantage of attackers, compared to the traditional encryption method, because the uplink data length is larger by $3N_t^*$, which improves security performance. Fig. 10 shows the advantage $\text{Adv}(\lambda)$ according to channel qualities and $\mathbb{P}[\beta = \beta']$. The advantage is high when $\mathbb{P}[\beta = \beta']$ is close to 0 or 1 and ρ has a high value (in high-BER environments), but the advantage is the lowest when $\mathbb{P}[\beta = \beta'] = 0.5$ regardless of channel qualities. That is, when the entropy of β is at the maximum, we can achieve the maximum security performance. Therefore, we should design $\mathcal{C}$ to have $\mathbb{P}[\beta = 0] = \mathbb{P}[\beta = 1] = 0.5$, which enhances the security performance by avoiding the impact of channel qualities.

E. Latency

Satellite communication systems have an inherent latency problem due to the physical distance from the ground. In this section, we analyze the total expected latency of the CFB-AES-TURBO method.

The probabilities of packet transmission success for downlink P^d and for uplink P^u are expressed by

$$P^d = (1 - P_e^d)^{N_t^d} \tag{37}$$

$$P^u = (1 - P_e^u)^{N_t^u} \tag{38}$$

where N_t^d and N_t^u are the EBSs of the respective links. We again assume independent bit errors in both (37) and (38). Here, we take into account N_t^d and N_t^u for data length because receivers consider only whether the decrypted data have errors or not. Each packet transmission over down/uplink is assumed to satisfy an independent and identically distributed Bernoulli trial [46]. Since satellite receives respond with acknowledgement (ACK) or negative ACK after data transmissions, total latency should be considered in terms of the BERs and EBSs of down/uplink. The numbers for successful transmissions for downlink N_{sd} and uplink N_{su} follow geometric distributions [47], and their averages $E[N_{\text{sd}}$ and $E[N_{\text{su}}]$ are given as

$$E[N_{\text{sd}}] = \frac{1}{P^d} \tag{39}$$

$$E[N_{\text{su}}] = \frac{1}{P^u}. \tag{40}$$

With the round-trip time (RTT) of each orbit satellite and average number of transmissions of (39) and (40), thus, the total expected latency L is calculated as

$$\begin{aligned} L &= \text{RTT} \cdot \Big(E[N_{\text{sd}}] + E[N_{\text{su}}]\Big) \\ &= \text{RTT} \cdot \frac{(1 - P_e^d)^{N_t^d} + (1 - P_e^u)^{N_t^u}}{\eta} \end{aligned} \tag{41}$$

where $\eta = (1 - P_e^d)^{N_t^d} \times (1 - P_e^u)^{N_t^u}$ denotes the probability of round-trip packet transmission success. Fig. 11 shows that the total expected latency of the CFB-AES-TURBO method for geostationary, medium, and low Earth orbits (GEO, MEO, and LEO, respectively) satellites. Here, L increases when η decreases and the altitude increases. The red asterisk marks indicate the best cases of the total expected latency when down/uplink conditions are symmetric with identical BERs. On the other hand, in the asymmetric case, since the increased number of retransmissions due to high BER (worse channel conditions) is much larger

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than the decreased number due to low BER (better channel conditions), the total expected latency rises exponentially.

As GEO satellites can communicate for 24 h with receivers, a new encryption key pair requires periodic updates. However, MEO and LEO satellites agree on the key pair, and use it with receivers when they enter the range of management of new receivers. Therefore, we set the total expected latency in (41) as the minimum key refresh time of GEO satellites, which means that a new encryption key pair agreement is executed after at least the total expected latency.

In summary, this section has shown that the proposed CFB-AES-TURBO method provides a short processing time by exploiting joint encryption and channel coding, simultaneously. The CFB-AES-TURBO method achieves a low BER of about 10^{-7} in the 4-dB SNR with a large EBS in the Rician fading satellite channel. Our method achieves robust and reliable security performance against brute-force and meet-in-the-middle attacks by using a different key pair for each block. Indistinguishability can be provided against the adversary, and the random design in data is strongly suggested. The latency analysis showed that the poor and asymmetric channel conditions deteriorate the total latency severely. In addition, the minimum key refresh time was proposed based on the total latency. We expect that the proposed method can be utilized for the security of the automatic dependent surveillance–broadcast system, with which aircrafts periodically broadcast their position information to be traced [48].

V. CONCLUSION

In this article, we proposed a novel joint method of CFB-AES-TURBO with a key distribution solution. By providing the capability of correcting bit errors and robustness against cryptanalysis, the proposed method can enhance the security of satellite data, which is exposed to physical layer attacks and bit errors in satellite channels. The CFB-AES-TURBO combines AES with turbo codes and the CFB mode to select the EBS, and provides a key exchange procedure. We derived processing time gains compared with conventional schemes and analyzed computational complexity of the proposed method, and then showed BERs of the received and decrypted data according to EBSs and channel conditions. Moreover, we achieved enhanced security performance against cryptanalysis of brute-force and meet-in-the middle attacks and provided indistinguishability, which can satisfy the practical requirements of BER and security. Finally, we analyzed the total expected latency of the CFB-AES-TURBO method for different orbits, and proposed the minimum key refresh time of the GEO satellites.

In practice, encryption of all the messages can be a big burden for the power-limited satellite system, by deteriorating the system efficiency. New integrated security solutions are required and being studied [49], [50]. Future work may include a multibeam satellite security method by a joint use of cryptographic security and PLS to improve the system efficiency of the future nonterrestrial network and space–air–terrestrial integrated network.

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